

Ross Enriquez

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Education

B.Eng., Computer Engineering, Toronto Metropolitan University

• **Academic Honors:** Dean's List (2019 - 2025)

GPA: 4.13/4.33

• **Relevant Courses:** Software Design and Architecture, Operating Systems, Database Systems, Data Structures and Algorithms, Object-Oriented Eng Analysis and Design, Computer Networks, and Software Project Management.

Skills

Languages: Java, Python, TypeScript, SQL, C

Frontend/UI: ReactJS, Next.js, Redux, ChakraUI, HTML/CSS/SASS

Backend & Tools: Node.js, Express.js, Flask, MongoDB, MySQL, Firebase

Cloud & DevOps: Google Cloud Platform, Docker, Kubernetes, Jenkins, Git, GitLab, CI/CD

Monitoring: Grafana, Graylog, Elasticsearch, Jira

Concepts: Microservices, Fault Tolerance, Production Debugging, Agile, Object-Oriented Design, JWT Auth

Work Experience

Software Developer

May 2023 - Dec 2024

Environment and Climate Change Canada

North York, ON

- Delivered object-oriented **Java** backend APIs with 50+ unit and integration tests (**JUnit**, **Mockito**), improving application reliability and maintainability
- Maintained and optimized 30+ taxonomy data sets across **PostgreSQL** and **Oracle** using custom **SQL** queries
- Integrated **Elasticsearch** to enhance system issue identification contributing to faster debugging
- Monitored distributed microservices via **Grafana** and **Graylog**, reducing incident detection time
- Maintained **CI/CD** pipelines with **Jenkins** for seamless production and staging deployments
- Participated in **Agile** ceremonies and cross-functional team collaboration; contributed to code reviews, technical documentation, and stakeholder alignment
- Provided system support, implementation assistance, and root cause analysis for bugs and production issues across staging and production environments

Projects

Speaker Diarization System for Dual-Speaker Audio - Python, Flask, PyTorch, WhisperX

[Demo](#)

- Led the design and implementation of an end-to-end speaker diarization application with real-time and pre-recorded processing pipelines using client-server architecture (**Flask**, **WebSockets**)
- Engineered audio preprocessing pipeline with **SileroVAD**, normalization, resampling (**PyTorch**), and speech chunk aggregation to optimize model performance and system latency
- Integrated **AssemblyAI** and **WhisperX** for transcription and diarization; designed local Resemblyzer-based speaker embedding calibration with cosine similarity for real-time speaker identification
- Performed extensive debugging, qualitative testing, and backend/frontend integration to ensure real-time transcription accuracy and responsiveness

Volleyball Drop-In Scheduler Bot - Python, discord.py, Firebase, Heroku

[Github](#)

- Developed a community Discord bot deployed on **Heroku** using **discord.py** and **Firebase**
- Implemented asynchronous request handling using Python's **asyncio**
- Handled event scheduling, stat collection, and dynamic queries for 50+ users

Car Rental Web App - Java, Spring Boot, Docker, Kubernetes, MySQL, Google Cloud, KubeMQ

[Github](#)

- Designed and developed a full-stack cloud-native car rental system with **REST APIs** (**Spring Boot**), secured authentication (**JWT**), and microservices messaging (**KubeMQ**)
- Deployed scalable backend services on GCP using **Docker** and **Kubernetes** for high uptime and performance
- Managed data with **MySQL** and optimized performance with indexing strategies

INDILATOR: Indigenous Language Translator - TypeScript, NextJS, MongoDB

[Github](#)

- Built a full-stack web translator app using **React**, **Next.js**, and **TypeScript** with multilingual support; led a 3-person team and placed 2nd at Ryerson Engineering Competition